16. Suppose that r_1 and r_2 are roots of $ar^2 + br + c = 0$ and that $r_1 \neq r_2$; then $\exp(r_1 t)$ and $\exp(r_2 t)$ are solutions of the differential equation $ay'' + by' + cy = 0$. Show that

$$\phi(t; r_1, r_2) = \frac{e^{r_2 t} - e^{r_1 t}}{r_2 - r_1}$$

is also a solution of the equation for $r_2 \neq r_1$. Then think of r_1 as fixed, and use l'Hôpital's rule to evaluate the limit of $\phi(t; r_1, r_2)$ as $r_2 \to r_1$, thereby obtaining the second solution in the case of equal roots.

17. **a.** If $ar^2 + br + c = 0$ has equal roots r_1, show that

$$L\left[e^{rt}\right] = a(e^{rt})'' + b(e^{rt})' + ce^{rt} = a(r - r_1)^2 e^{rt}. \quad (37)$$

Since the right-hand side of equation (37) is zero when $r = r_1$, it follows that $\exp(r_1 t)$ is a solution of $L[y] = ay'' + by' + cy = 0$.

b. Differentiate equation (37) with respect to r, and interchange differentiation with respect to r and with respect to t, thus showing that

$$\frac{\partial}{\partial r} L\left[e^{rt}\right] = L\left[\frac{\partial}{\partial r} e^{rt}\right] = L\left[te^{rt}\right]$$
$$= ate^{rt}(r - r_1)^2 + 2ae^{rt}(r - r_1). \quad (38)$$

Since the right-hand side of equation (36) is zero when $r = r_1$, conclude that $t \exp(r_1 t)$ is also a solution of $L[y] = 0$.

In each of Problems 18 through 22, use the method of reduction of order to find a second solution of the given differential equation.

18. $t^2 y'' - 4ty' + 6y = 0, \quad t > 0; \quad y_1(t) = t^2$

19. $t^2 y'' + 2ty' - 2y = 0, \quad t > 0; \quad y_1(t) = t$

20. $t^2 y'' + 3ty' + y = 0, \quad t > 0; \quad y_1(t) = t^{-1}$

21. $xy'' - y' + 4x^3 y = 0, \quad x > 0; \quad y_1(x) = \sin(x^2)$

22. $x^2 y'' + xy' + (x^2 - 0.25)y = 0, \quad x > 0; \quad y_1(x) = x^{-1/2} \sin x$

23. The differential equation

$$y'' + \delta(xy' + y) = 0$$

arises in the study of the turbulent flow of a uniform stream past a circular cylinder. Verify that $y_1(x) = \exp(-\delta x^2/2)$ is one solution, and then find the general solution in the form of an integral.

24. The method of Problem 15 can be extended to second-order equations with variable coefficients. If y_1 is a known nonvanishing solution of $y'' + p(t)y' + q(t)y = 0$, show that a second solution y_2 satisfies $(y_2/y_1)' = W[y_1, y_2]/y_1^2$, where $W[y_1, y_2]$ is the Wronskian of y_1 and y_2. Then use Abel's formula (equation (23) of Section 3.2) to determine y_2.

In each of Problems 25 through 27, use the method of Problem 24 to find a second independent solution of the given equation.

25. $t^2 y'' + 3ty' + y = 0, \quad t > 0; \quad y_1(t) = t^{-1}$

26. $ty'' - y' + 4t^3 y = 0, \quad t > 0; \quad y_1(t) = \sin(t^2)$

27. $x^2 y'' + xy' + (x^2 - 0.25)y = 0, \quad x > 0; \quad y_1(x) = x^{-1/2} \sin x$

Behavior of Solutions as $t \to \infty$. Problems 28 through 30 are concerned with the behavior of solutions as $t \to \infty$.

28. If a, b, and c are positive constants, show that all solutions of $ay'' + by' + cy = 0$ approach zero as $t \to \infty$.

29. **a.** If $a > 0$ and $c > 0$, but $b = 0$, show that the result of Problem 28 is no longer true, but that all solutions are bounded as $t \to \infty$.

b. If $a > 0$ and $b > 0$, but $c = 0$, show that the result of Problem 28 is no longer true, but that all solutions approach a constant that depends on the initial conditions as $t \to \infty$. Determine this constant for the initial conditions $y(0) = y_0$, $y'(0) = y_0'$.

30. Show that $y = \sin t$ is a solution of

$$y'' + (k \sin^2 t)y' + (1 - k \cos t \sin t)y = 0$$

for any value of the constant k. If $0 < k < 2$, show that $1 - k \cos t \sin t > 0$ and $k \sin^2 t \geq 0$. Thus observe that even though the coefficients of this variable-coefficient differential equation are nonnegative (and the coefficient of y' is zero only at the points $t = 0, \pi, 2\pi, \ldots$), it has a solution that does not approach zero as $t \to \infty$. Compare this situation with the result of Problem 28. Thus we observe a not unusual situation in the study of differential equations: equations that are apparently very similar can have quite different properties.

Euler Equations. In each of Problems 31 through 34, use the substitution introduced in Problem 25 in Section 3.3 to solve the given differential equation.

31. $t^2 y'' - 3ty' + 4y = 0, \quad t > 0$

32. $t^2 y'' + 2ty' + 0.25y = 0, \quad t > 0$

33. $t^2 y'' + 3ty' + y = 0, \quad t > 0$

34. $4t^2 y'' - 8ty' + 9y = 0, \quad t > 0$

3.5 Nonhomogeneous Equations; Method of Undetermined Coefficients

We now turn our attention to the nonhomogeneous second-order linear differential equation

$$L[y] = y'' + p(t)y' + q(t)y = g(t), \quad (1)$$

where p, q, and g are given (continuous) functions on the open interval I. The equation

$$L[y] = y'' + p(t)y' + q(t)y = 0, \quad (2)$$

in which $g(t) = 0$ and p and q are the same as in equation (1), is called the homogeneous differential equation corresponding to equation (1). The following two results describe the structure of solutions of the nonhomogeneous equation (1) and provide a foundation for constructing its general solution.

Theorem 3.5.1

If Y_1 and Y_2 are two solutions of the nonhomogeneous linear differential equation (1), then their difference $Y_1 - Y_2$ is a solution of the corresponding homogeneous differential equation (2). If, in addition, y_1 and y_2 form a fundamental set of solutions of equation (2), then

$$Y_1(t) - Y_2(t) = c_1 y_1(t) + c_2 y_2(t), \tag{3}$$

where c_1 and c_2 are certain constants.

To prove this result, note that Y_1 and Y_2 satisfy the equations

$$L[Y_1](t) = g(t), \quad L[Y_2](t) = g(t). \tag{4}$$

Subtracting the second of these equations from the first, we have

$$L[Y_1](t) - L[Y_2](t) = g(t) - g(t) = 0. \tag{5}$$

However,

$$L[Y_1] - L[Y_2] = L[Y_1 - Y_2],$$

so equation (5) becomes

$$L[Y_1 - Y_2](t) = 0. \tag{6}$$

Equation (6) states that $Y_1 - Y_2$ is a solution of equation (2). Finally, since by Theorem 3.2.4 all solutions of equation (2) can be expressed as linear combinations of a fundamental set of solutions, it follows that the solution $Y_1 - Y_2$ can be so written. Hence equation (3) holds and the proof is complete.

Theorem 3.5.2

The general solution of the nonhomogeneous equation (1) can be written in the form

$$y = \phi(t) = c_1 y_1(t) + c_2 y_2(t) + Y(t), \tag{7}$$

where y_1 and y_2 form a fundamental set of solutions of the corresponding homogeneous equation (2), c_1 and c_2 are arbitrary constants, and Y is any solution of the nonhomogeneous equation (1).

The proof of Theorem 3.5.2 follows quickly from Theorem 3.5.1. Note that equation (3) holds if we identify Y_1 with an arbitrary solution ϕ of equation (1) and Y_2 with the specific solution Y. From equation (3) we thereby obtain

$$\phi(t) - Y(t) = c_1 y_1(t) + c_2 y_2(t), \tag{8}$$

which is equivalent to equation (7). Since ϕ is an arbitrary solution of equation (1), the expression on the right-hand side of equation (7) includes all solutions of equation (1); thus it is natural to call it the general solution of equation (1).

In somewhat different words, Theorem 3.5.2 states that to solve the nonhomogeneous equation (1), we must do three things:

1. Find the general solution $c_1 y_1(t) + c_2 y_2(t)$ of the corresponding homogeneous equation. This solution is frequently called the **complementary solution** and may be denoted by $y_c(t)$.
2. Find any solution $Y(t)$ of the nonhomogeneous equation. Often this solution is referred to as a **particular solution**.
3. Form the sum of the functions found in steps 1 and 2.

We have already discussed how to find $y_c(t)$, at least when the homogeneous equation (2) has constant coefficients. Therefore, in the remainder of this section and Section 3.6, we

will focus on finding a particular solution $Y(t)$ of the nonhomogeneous linear differential equation (1). There are two methods that we wish to discuss. They are known as the method of undetermined coefficients (discussed here) and the method of variation of parameters (see Section 3.6). Each has some advantages and some possible shortcomings.

Method of Undetermined Coefficients. The method of undetermined coefficients requires us to make an initial assumption about the form of the particular solution $Y(t)$, but with the coefficients left unspecified. We then substitute the assumed expression into the nonhomogeneous differential equation (1) and attempt to determine the coefficients so as to satisfy that equation. If we are successful, then we have found a solution of the differential equation (1) and can use it for the particular solution $Y(t)$. If we cannot determine the coefficients, then this means that there is no solution of the form that we assumed. In this case we may modify the initial assumption and try again.

The main advantage of the method of undetermined coefficients is that it is straightforward to execute once the assumption is made about the form of $Y(t)$. Its major limitation is that it is useful primarily for equations for which we can easily write down the correct form of the particular solution in advance. For this reason, this method is usually used only for problems in which the homogeneous equation has constant coefficients and the nonhomogeneous term is restricted to a relatively small class of functions. In particular, we consider only nonhomogeneous terms that consist of polynomials, exponential functions, sines, and cosines. Despite this limitation, the method of undetermined coefficients is useful for solving many problems that have important applications. However, the algebraic details may become tedious, and a computer algebra system can be very helpful in practical applications. We will illustrate the method of undetermined coefficients by several simple examples and then summarize some rules for using it.

EXAMPLE 1

Find a particular solution of

$$y'' - 3y' - 4y = 3e^{2t}. \tag{9}$$

Solution:

We seek a function Y such that the combination $Y''(t) - 3Y'(t) - 4Y(t)$ is equal to $3e^{2t}$. Since the exponential function reproduces itself through differentiation, the most plausible way to achieve the desired result is to assume that $Y(t)$ is some multiple of e^{2t},

$$Y(t) = Ae^{2t},$$

where the coefficient A is yet to be determined. To find A, we calculate the first two derivatives of Y:

$$Y'(t) = 2Ae^{2t}, \quad Y''(t) = 4Ae^{2t},$$

and substitute for y, y', and y'' in the nonhomogeneous differential equation (9). We obtain

$$Y'' - 3Y' - 4Y = (4A - 6A - 4A)e^{2t} = 3e^{2t}.$$

Hence $-6Ae^{2t}$ must equal $3e^{2t}$, so $-6A = 3$ and we conclude that $A = -\frac{1}{2}$. Thus a particular solution is

$$Y(t) = -\frac{1}{2}e^{2t}. \tag{10}$$

EXAMPLE 2

Find a particular solution of

$$y'' - 3y' - 4y = 2\sin t. \tag{11}$$

Solution:

By analogy with Example 1, let us assume that $Y(t) = A\sin t$, where A is a constant to be determined. On substituting this guess in equation (11) we obtain

$$Y'' - 3Y' - 4Y = -A\sin t - 3A\cos t - 4A\sin t = 2\sin t,$$

or, moving all terms to the left-hand side and collecting the terms involving $\sin t$ and $\cos t$, we arrive at,

$$(2 + 5A)\sin t + 3A\cos t = 0. \tag{12}$$

We want equation (12) to hold for all t. Thus it must hold for two specific points, such as $t = 0$ and $t = \frac{\pi}{2}$. At these points equation (12) reduces to $3A = 0$ and $2 + 5A = 0$, respectively. These contradictory requirements mean that there is no choice of the constant A that makes equation (12) true for $t = 0$ and $t = \frac{\pi}{2}$, much less for all t. Thus we conclude that our assumption concerning $Y(t)$ is inadequate.

The appearance of the cosine term in equation (12) suggests that we modify our original assumption to include a cosine term in $Y(t)$; that is,

$$Y(t) = A\sin t + B\cos t,$$

where A and B are the undetermined coefficients. Then

$$Y'(t) = A\cos t - B\sin t, \quad Y''(t) = -A\sin t - B\cos t.$$

By substituting these expressions for y, y', and y'' in equation (11) and collecting terms, we obtain

$$Y'' - 3Y' - 4Y = (-A + 3B - 4A)\sin t + (-B - 3A - 4B)\cos t = 2\sin t. \tag{13}$$

Now, working exactly as with the first guess, move all terms to the left-hand side and evaluate $t = 0$ and $t = \frac{\pi}{2}$ to find that A and B must satisfy the equations

$$-5A + 3B - 2 = 0, \quad -3A - 5B = 0.$$

Solving these algebraic equations for A and B, we obtain $A = -\frac{5}{17}$ and $B = \frac{3}{17}$; hence a particular solution of equation (11) is

$$Y(t) = -\frac{5}{17}\sin t + \frac{3}{17}\cos t.$$

The method illustrated in the preceding examples can also be used when the right-hand side of the equation is a polynomial. Thus, to find a particular solution of

$$y'' - 3y' - 4y = 4t^2 - 1, \tag{14}$$

we initially assume that $Y(t)$ is a polynomial of the same degree as the nonhomogeneous term; that is, $Y(t) = At^2 + Bt + C$.

To summarize our conclusions up to this point: if the nonhomogeneous term $g(t)$ in differential equation (1) is an exponential function $e^{\alpha t}$, then assume that $Y(t)$ is proportional to the same exponential function; if $g(t)$ is $\sin(\beta t)$ or $\cos(\beta t)$, then assume that $Y(t)$ is a linear combination of $\sin(\beta t)$ and $\cos(\beta t)$; if $g(t)$ is a polynomial of degree n, then assume that $Y(t)$ is a polynomial of degree n. The same principle extends to the case where $g(t)$ is a product of any two, or all three, of these types of functions, as the next example illustrates.

EXAMPLE 3

Find a particular solution of

$$y'' - 3y' - 4y = -8e^t\cos(2t). \tag{15}$$

Solution:

In this case we assume that $Y(t)$ is the product of e^t and a linear combination of $\cos(2t)$ and $\sin(2t)$; that is,

$$Y(t) = Ae^t \cos(2t) + Be^t \sin(2t).$$

The algebra is more tedious in this example, but it follows that

$$Y'(t) = (A + 2B)e^t \cos(2t) + (-2A + B)e^t \sin(2t)$$

and

$$Y''(t) = (-3A + 4B)e^t \cos(2t) + (-4A - 3B)e^t \sin(2t).$$

By substituting these expressions in equation (15), we find that A and B must satisfy

$$10A + 2B = 8, \quad 2A - 10B = 0.$$

Hence $A = \frac{10}{13}$ and $B = \frac{2}{13}$; therefore, a particular solution of equation (15) is

$$Y(t) = \frac{10}{13}e^t \cos(2t) + \frac{2}{13}e^t \sin(2t).$$

Now suppose that $g(t)$ is the sum of two terms, $g(t) = g_1(t) + g_2(t)$, and suppose that Y_1 and Y_2 are solutions of the equations

$$ay'' + by' + cy = g_1(t) \tag{16}$$

and

$$ay'' + by' + cy = g_2(t), \tag{17}$$

respectively. Then $Y_1 + Y_2$ is a solution of the equation

$$ay'' + by' + cy = g(t). \tag{18}$$

To prove this statement, substitute $Y_1(t) + Y_2(t)$ for y in equation (18) and make use of equations (16) and (17). A similar conclusion holds if $g(t)$ is the sum of any finite number of terms. The practical significance of this result is that for an equation whose nonhomogeneous function $g(t)$ can be expressed as a sum, you can consider instead several simpler equations and then add together the results. The following example is an illustration of this procedure.

EXAMPLE 4

Find a particular solution of

$$y'' - 3y' - 4y = 3e^{2t} + 2\sin t - 8e^t \cos(2t). \tag{19}$$

Solution:

By splitting up the right-hand side of equation (19), we obtain the three equations

$$y'' - 3y' - 4y = 3e^{2t},$$
$$y'' - 3y' - 4y = 2\sin t,$$

and

$$y'' - 3y' - 4y = -8e^t \cos(2t).$$

Solutions of these three equations have been found in Examples 1, 2, and 3, respectively. Therefore, a particular solution of equation (19) is their sum, namely,

$$Y(t) = -\frac{1}{2}e^{2t} + \frac{3}{17}\cos t - \frac{5}{17}\sin t + \frac{10}{13}e^t \cos(2t) + \frac{2}{13}e^t \sin(2t).$$

The procedure illustrated in these examples enables us to solve a fairly large class of problems in a reasonably efficient manner. However, there is one difficulty that sometimes occurs. The next example illustrates how it arises.

EXAMPLE 5

Find a particular solution of

$$y'' - 3y' - 4y = 2e^{-t}. \tag{20}$$

Solution:

Proceeding as in Example 1, we assume that $Y(t) = Ae^{-t}$. By substituting in equation (20), we obtain

$$Y'' - 3Y' - 4Y = (A + 3A - 4A)e^{-t} = 2e^{-t}. \tag{21}$$

Since the left-hand side of equation (21) is zero, there is no choice of A for which $0 = 2e^{-t}$. Therefore, there is no particular solution of equation (20) of the assumed form. The reason for this possibly unexpected result becomes clear if we solve the homogeneous equation

$$y'' - 3y' - 4y = 0 \tag{22}$$

that corresponds to equation (20). The two functions in a fundamental set of solutions of equation (22) are $y_1(t) = e^{-t}$ and $y_2(t) = e^{4t}$. Thus our assumed particular solution of equation (20) is actually a solution of the homogeneous equation (22); consequently, it cannot possibly be a solution of the nonhomogeneous equation (20). To find a solution of equation (20), we must therefore consider functions of a somewhat different form.

At this stage, we have several possible alternatives. One is simply to try to guess the proper form of the particular solution of equation (20). Another is to solve this equation in some different way and then to use the result to guide our assumptions if this situation arises again in the future; see Problems 22 and 27 for other solution methods. Still another possibility is to seek a simpler equation where this difficulty occurs and to use its solution to suggest how we might proceed with equation (20). Adopting the latter approach, we look for a first-order equation analogous to equation (20). One possibility is the linear equation

$$y' + y = 2e^{-t}. \tag{23}$$

If we try to find a particular solution of equation (23) of the form Ae^{-t}, we will fail because e^{-t} is a solution of the homogeneous equation $y' + y = 0$. However, from Section 2.1 we already know how to solve equation (23). An integrating factor is $\mu(t) = e^{t}$, and by multiplying by $\mu(t)$ and then integrating both sides, we obtain the solution

$$y = 2te^{-t} + ce^{-t}. \tag{24}$$

The second term on the right-hand side of equation (24) is the general solution of the homogeneous equation $y' + y = 0$, but the first term is a solution of the full nonhomogeneous equation (23). Observe that it involves the exponential factor e^{-t} multiplied by the factor t. This is the clue that we were looking for.

We now return to equation (20) and assume a particular solution of the form $Y(t) = Ate^{-t}$. Then

$$Y'(t) = Ae^{-t} - Ate^{-t}, \quad Y''(t) = -2Ae^{-t} + Ate^{-t}. \tag{25}$$

Substituting these expressions for y, y', and y'' in equation (20), we obtain

$$Y'' - 3Y' - 4Y = (-2A - 3A)e^{-t} + (A + 3A - 4A)te^{-t} = 2e^{-t}.$$

The coefficient of te^{-t} is zero, and from the terms involving e^{t} we have $-5A = 2$, so $A = -\frac{2}{5}$. Thus a particular solution of equation (20) is

$$Y(t) = -\frac{2}{5}te^{-t}. \tag{26}$$

The outcome of Example 5 suggests a modification of the principle stated previously: if the assumed form of the particular solution duplicates a solution of the corresponding homogeneous equation, then modify the assumed particular solution by multiplying it by t. Occasionally, this modification will be insufficient to remove all duplication with the solutions of the homogeneous equation, in which case it is necessary to multiply by t a second time. For a second-order equation, it will never be necessary to carry the process further than this.

Summary. We now summarize the steps involved in finding the solution of an initial value problem consisting of a nonhomogeneous linear differential equation of the form

$$ay'' + by' + cy = g(t), \tag{27}$$

where the coefficients a, b, and c are constants, together with a given set of initial conditions.

1. Find the general solution of the corresponding homogeneous equation.
2. Make sure that the function $g(t)$ in equation (27) belongs to the class of functions discussed in this section; that is, be sure it involves nothing more than exponential functions, sines, cosines, polynomials, or sums or products of such functions. If this is not the case, use the method of variation of parameters (discussed in Section 3.6).
3. If $g(t) = g_1(t) + \cdots + g_n(t)$—that is, if $g(t)$ is a sum of n terms—then form n subproblems, each of which contains only one of the terms $g_1(t), \ldots, g_n(t)$. The ith subproblem consists of the equation
$$ay'' + by' + cy = g_i(t),$$
where i runs from 1 to n.
4. For the ith subproblem, assume a particular solution $Y_i(t)$ consisting of the appropriate exponential function, sine, cosine, polynomial, or combination thereof. If there is any duplication in the assumed form of $Y_i(t)$ with the solutions of the homogeneous equation (found in step 1), then multiply $Y_i(t)$ by t, or (if necessary) by t^2, so as to remove the duplication. See Table 3.5.1.

TABLE 3.5.1 The Particular Solution of $ay'' + by' + cy = g_i(t)$

$g_i(t)$	$Y_i(t)$
$P_n(t) = a_0t^n + a_1t^{n-1} + \cdots + a_n$	$t^s(A_0t^n + A_1t^{n-1} + \cdots + A_n)$
$P_n(t)e^{\alpha t}$	$t^s(A_0t^n + A_1t^{n-1} + \cdots + A_n)e^{\alpha t}$
$P_n(t)e^{\alpha t}\begin{cases}\sin\beta t\\ \cos\beta t\end{cases}$	$t^s\big((A_0t^n + A_1t^{n-1} + \cdots + A_n)e^{\alpha t}\cos(\beta t)$ $+(B_0t^n + B_1t^{n-1} + \cdots + B_n)e^{\alpha t}\sin(\beta t)\big)$

Notes: Here, s is the smallest nonnegative integer ($s = 0$, 1, or 2) that will ensure that no term in $Y_i(t)$ is a solution of the corresponding homogeneous equation. Equivalently, for the three cases, s is the number of times 0 is a root of the characteristic equation, α is a root of the characteristic equation, and $\alpha + i\beta$ is a root of the characteristic equation, respectively.

5. Find a particular solution $Y_i(t)$ for each of the subproblems. Then $Y_1(t) + \cdots + Y_n(t)$ is a particular solution of the full nonhomogeneous equation (27).
6. Form the sum of the general solution of the homogeneous equation (step 1) and the particular solution of the nonhomogeneous equation (step 5). This is the general solution of the nonhomogeneous equation.
7. When initial conditions are provided, use them to determine the values of the arbitrary constants remaining in the general solution.

For some problems this entire procedure is easy to carry out by hand, but often the algebraic calculations are lengthy. Once you understand clearly how the method works, a computer algebra system can be of great assistance in executing the details.

The method of undetermined coefficients is self-correcting in the sense that if you assume too little for $Y(t)$, then a contradiction is soon reached that usually points the way to the modification that is needed in the assumed form. On the other hand, if you assume too many terms, then some unnecessary work is done and some coefficients turn out to be zero, but at least the correct answer is obtained.

Proof of the Method of Undetermined Coefficients. In the preceding discussion we have described the method of undetermined coefficients on the basis of several examples. To prove that the procedure always works as stated, we now give a general argument, in which we consider three cases corresponding to different forms for the nonhomogeneous term $g(t)$.

Case 1: $g(t) = P_n(t) = a_0t^n + a_1t^{n-1} + \cdots + a_n$. In this case equation (27) becomes
$$ay'' + by' + cy = a_0t^n + a_1t^{n-1} + \cdots + a_n. \tag{28}$$

To obtain a particular solution, we assume that

$$Y(t) = A_0t^n + A_1t^{n-1} + \cdots + A_{n-2}t^2 + A_{n-1}t + A_n. \tag{29}$$

Substituting in equation (28), we obtain

$$a\Big(n(n-1)A_0t^{n-2} + \cdots + 2A_{n-2}\Big) + b(nA_0t^{n-1} + \cdots + A_{n-1}) + c(A_0t^n + A_1t^{n-1} + \cdots + A_n) = a_0t^n + \cdots + a_n. \tag{30}$$

Equating the coefficients of like powers of t, beginning with t^n, leads to the following sequence of equations:

$$\begin{aligned} cA_0 &= a_0, \\ cA_1 + nbA_0 &= a_1, \\ &\vdots \\ cA_n + bA_{n-1} + 2aA_{n-2} &= a_n. \end{aligned}$$

Provided that $c \neq 0$, the solution of the first equation is $A_0 = a_0/c$, and the remaining equations determine $A_1, \ldots, A_n$ successively.

If $c = 0$ but $b \neq 0$, then the polynomial on the left-hand side of equation (30) is of degree $n-1$, and we cannot satisfy equation (30). To be sure that $aY''(t) + bY'(t)$ is a polynomial of degree n, we must choose $Y(t)$ to be a polynomial of degree $n+1$. Hence we assume that

$$Y(t) = t(A_0t^n + \cdots + A_n).$$

Substituting this guess into equation (28), with $c = 0$, and simplifying yields

$$\begin{aligned} aY'' + bY' &= bA_0(n+1)t^n + (aA_0(n+1)n + bA_1n)t^{n-1} + \cdots \\ &= a_0t^n + a_1t^{n-1} + \cdots + a_n. \end{aligned}$$

There is no constant term in this expression for $Y(t)$, but there is no need to include such a term since a constant is a solution of the homogeneous equation when $c = 0$. Since $b \neq 0$, we find $A_0 = a_0/(b(n+1))$, and the other coefficients $A_1, \ldots, A_n$ can be determined similarly.

If both c and b are zero, then the characteristic equation is $ar^2 = 0$ and $r = 0$ is a repeated root. Thus $y_1 = e^{0t} = 1$ and $y_2 = te^{0t} = t$ form a fundamental set of solutions of the corresponding homogeneous equation. This leads us to assume that

$$Y(t) = t^2(A_0t^n + \cdots + A_n).$$

The term $aY''(t)$ gives rise to a term of degree n, and we can proceed as before. Again the constant and linear terms in $Y(t)$ are omitted since, in this case, they are both solutions of the homogeneous equation.

Case 2: $g(t) = e^{\alpha t}P_n(t)$. The problem of determining a particular solution of

$$ay'' + by' + cy = e^{\alpha t}P_n(t) \tag{31}$$

can be reduced to the preceding case by a substitution. Let

$$Y(t) = e^{\alpha t}u(t);$$

then

$$Y'(t) = e^{\alpha t}(u'(t) + \alpha u(t))$$

and

$$Y''(t) = e^{\alpha t}(u''(t) + 2\alpha u'(t) + \alpha^2 u(t)).$$

Substituting for y, y', and y'' in equation (31), canceling the factor $e^{\alpha t}$, and collecting terms, we obtain

$$au''(t) + (2a\alpha + b)u'(t) + (a\alpha^2 + b\alpha + c)u(t) = P_n(t). \tag{32}$$

The determination of a particular solution of equation (32) is precisely the same problem, except for the names of the constants, as solving equation (28). Therefore, if $a\alpha^2 + b\alpha + c$ is not zero, we assume that $u(t) = A_0t^n + \cdots + A_n$; hence a particular solution of equation (31) is of the form

$$Y(t) = e^{\alpha t}(A_0t^n + A_1t^{n-1} + \cdots + A_n). \tag{33}$$

On the other hand, if $a\alpha^2 + b\alpha + c$ is zero but $2a\alpha + b$ is not, we must take $u(t)$ to be of the form $t(A_0t^n + \cdots + A_n)$. The corresponding form for $Y(t)$ is t times the expression on the right-hand side of equation (33). Note that if $a\alpha^2 + b\alpha + c$ is zero, then $e^{\alpha t}$ is a solution of the homogeneous equation.

If both $a\alpha^2 + b\alpha + c$ and $2a\alpha + b$ are zero (and this implies that both $e^{\alpha t}$ and $te^{\alpha t}$ are solutions of the homogeneous equation), then the correct form for $u(t)$ is $t^2(A_0t^n + \cdots + A_n)$. Hence $Y(t)$ is t^2 times the expression on the right-hand side of equation (33).

Case 3: $g(t) = e^{\alpha t}P_n(t)\cos(\beta t)$ or $e^{\alpha t}P_n(t)\sin(\beta t)$. These two cases are similar, so we consider only the latter. We can reduce this problem to the preceding one by noting that, as a consequence of Euler's formula, $\sin(\beta t) = (e^{i\beta t} - e^{-i\beta t})/(2i)$. Hence $g(t)$ is of the form

$$g(t) = P_n(t)\frac{e^{(\alpha+i\beta)t} - e^{(\alpha-i\beta)t}}{2i},$$

and we should choose

$$Y(t) = e^{(\alpha+i\beta)t}(A_0t^n + \cdots + A_n) + e^{(\alpha-i\beta)t}(B_0t^n + \cdots + B_n),$$

or, equivalently,

$$Y(t) = e^{\alpha t}(A_0t^n + \cdots + A_n)\cos(\beta t) + e^{\alpha t}(B_0t^n + \cdots + B_n)\sin(\beta t).$$

Usually, the latter form is preferred because it does not involve the use of complex-valued coefficients. If $\alpha \pm i\beta$ satisfy the characteristic equation corresponding to the homogeneous equation, we must, of course, multiply each of the polynomials by t to increase their degrees by 1.

If the nonhomogeneous function involves both $\cos(\beta t)$ and $\sin(\beta t)$, it is usually convenient to treat these terms together, since each one individually may give rise to the same form for a particular solution. For example, if $g(t) = t\sin t + 2\cos t$, the form for $Y(t)$ would be

$$Y(t) = (A_0t + A_1)\sin t + (B_0t + B_1)\cos t,$$

provided that $\sin t$ and $\cos t$ are not solutions of the homogeneous equation.

Problems

In each of Problems 1 through 10, find the general solution of the given differential equation.

1. $y'' - 2y' - 3y = 3e^{2t}$

2. $y'' - y' - 2y = -2t + 4t^2$

3. $y'' + y' - 6y = 12e^{3t} + 12e^{-2t}$

4. $y'' - 2y' - 3y = -3te^{-t}$

5. $y'' + 2y' = 3 + 4\sin(2t)$

6. $y'' + 2y' + y = 2e^{-t}$

7. $y'' + y = 3\sin(2t) + t\cos(2t)$

8. $u'' + \omega_0^2 u = \cos(\omega t), \quad \omega^2 \neq \omega_0^2$

9. $u'' + \omega_0^2 u = \cos(\omega_0 t)$

10. $y'' + y' + 4y = 2\sinh t$ *Hint:* $\sinh t = (e^t - e^{-t})/2$

In each of Problems 11 through 15, find the solution of the given initial value problem.

11. $y'' + y' - 2y = 2t, \quad y(0) = 0, \quad y'(0) = 1$

12. $y'' + 4y = t^2 + 3e^t, \quad y(0) = 0, \quad y'(0) = 2$

13. $y'' - 2y' + y = te^t + 4, \quad y(0) = 1, \quad y'(0) = 1$

14. $y'' + 4y = 3\sin(2t), \quad y(0) = 2, \quad y'(0) = -1$

15. $y'' + 2y' + 5y = 4e^{-t}\cos(2t), \quad y(0) = 1, \quad y'(0) = 0$

In each of Problems 16 through 21:

a. Determine a suitable form for $Y(t)$ if the method of undetermined coefficients is to be used.

N b. Use a computer algebra system to find a particular solution of the given equation.

16. $y'' + 3y' = 2t^4 + t^2e^{-3t} + \sin(3t)$

17. $y'' - 5y' + 6y = e^t\cos(2t) + e^{2t}(3t + 4)\sin t$

18. $y'' + 2y' + 2y = 3e^{-t} + 2e^{-t}\cos t + 4e^{-t}t^2\sin t$

19. $y'' + 4y = t^2\sin(2t) + (6t + 7)\cos(2t)$

20. $y'' + 3y' + 2y = e^t(t^2 + 1)\sin(2t) + 3e^{-t}\cos t + 4e^t$

21. $y'' + 2y' + 5y = 3te^{-t}\cos(2t) - 2te^{-2t}\cos t$

22. Consider the equation

$$y'' - 3y' - 4y = 2e^{-t} \tag{34}$$

from Example 5. Recall that $y_1(t) = e^{-t}$ and $y_2(t) = e^{4t}$ are solutions of the corresponding homogeneous equation. Adapting the method of reduction of order (Section 3.4), seek a solution of the nonhomogeneous equation of the form $Y(t) = v(t)y_1(t) = v(t)e^{-t}$, where $v(t)$ is to be determined.

a. Substitute $Y(t)$, $Y'(t)$, and $Y''(t)$ into equation (34) and show that $v(t)$ must satisfy $v'' - 5v' = 2$.
b. Let $w(t) = v'(t)$ and show that $w(t)$ satisfies $w' - 5w = 2$. Solve this equation for $w(t)$.
c. Integrate $w(t)$ to find $v(t)$ and then show that

$$Y(t) = -\frac{2}{5}te^{-t} + \frac{1}{5}c_1e^{4t} + c_2e^{-t}.$$

The first term on the right-hand side is the desired particular solution of the nonhomogeneous equation. Note that it is a product of t and e^{-t}.

23. Determine the general solution of

$$y'' + \lambda^2 y = \sum_{m=1}^{N} a_m \sin(m\pi t),$$

where $\lambda > 0$ and $\lambda \neq m\pi$ for $m = 1, \ldots, N$.

N 24. In many physical problems the nonhomogeneous term may be specified by different formulas in different time periods. As an example, determine the solution $y = \phi(t)$ of

$$y'' + y = \begin{cases} t, & 0 \le t \le \pi, \\ \pi e^{\pi - t}, & t > \pi, \end{cases}$$

satisfying the initial conditions $y(0) = 0$ and $y'(0) = 1$. Assume that y and y' are also continuous at $t = \pi$. Plot the nonhomogeneous term and the solution as functions of time. *Hint:* First solve the initial value problem for $t \le \pi$; then solve for $t > \pi$, determining the constants in the latter solution from the continuity conditions at $t = \pi$.

Behavior of Solutions as $t \to \infty$. In Problems 25 and 26, we continue the discussion started with Problems 28 through 30 of Section 3.4. Consider the differential equation

$$ay'' + by' + cy = g(t), \tag{35}$$

where a, b, and c are positive.

25. If $Y_1(t)$ and $Y_2(t)$ are solutions of equation (35), show that $Y_1(t) - Y_2(t) \to 0$ as $t \to \infty$. Is this result true if $b = 0$?

26. If $g(t) = d$, a constant, show that every solution of equation (35) approaches d/c as $t \to \infty$. What happens if $c = 0$? What if $b = 0$ also?

27. In this problem we indicate an alternative procedure[8] for solving the differential equation

$$y'' + by' + cy = (D^2 + bD + c)y = g(t), \tag{36}$$

where b and c are constants, and D denotes differentiation with respect to t. Let r_1 and r_2 be the zeros of the characteristic polynomial of the corresponding homogeneous equation. These roots may be real and different, real and equal, or conjugate complex numbers.

a. Verify that equation (36) can be written in the factored form

$$(D - r_1)(D - r_2)y = g(t),$$

where $r_1 + r_2 = -b$ and $r_1r_2 = c$.
b. Let $u = (D - r_2)y$. Then show that the solution of equation (36) can be found by solving the following two first-order equations:

$$(D - r_1)u = g(t), \quad (D - r_2)y = u(t).$$

In each of Problems 28 through 30, use the method of Problem 27 to solve the given differential equation.

28. $y'' - 3y' - 4y = 3e^{2t}$ (see Example 1)
29. $y'' + 2y' + y = 2e^{-t}$ (see Problem 6)
30. $y'' + 2y' = 3 + 4\sin(2t)$ (see Problem 5)

[8] R. S. Luthar, "Another Approach to a Standard Differential Equation," *Two Year College Mathematics Journal 10* (1979), pp. 200–201. Also see D. C. Sandell and F. M. Stein, "Factorization of Operators of Second-Order Linear Homogeneous Ordinary Differential Equations," *Two Year College Mathematics Journal 8* (1977), pp. 132–141, for a more general discussion of factoring operators.

3.6 Variation of Parameters

In this section we describe a second method of finding a particular solution of a nonhomogeneous equation. This method, **variation of parameters**, is due to Lagrange and complements the method of undetermined coefficients rather well. The main advantage of variation of parameters is that it is a *general method*; in principle at least, it can be applied to any equation, and it requires no detailed assumptions about the form of the solution. In fact, later in this section we use this method to derive a formula for a particular solution of an arbitrary second-order linear nonhomogeneous differential equation. On the other hand, the method of variation of parameters eventually requires us to evaluate certain integrals involving the nonhomogeneous term in the differential equation, and this may present difficulties. Before looking at this method in the general case, we illustrate its use in an example.

EXAMPLE 1

Find the general solution of

$$y'' + 4y = 8\tan t \quad -\pi/2 < t < \pi/2. \tag{1}$$